

Belief Propagation-Based Polar Decoders With Schedule Diversity

Cheng-Jun You^{ID}, Yi-Hsin Chen, Chen-Wei Hu^{ID}, Huang-Chang Lee^{ID}, *Member, IEEE*,
and Yen-Ming Chen^{ID}, *Member, IEEE*

Abstract—Based on pruning techniques that interpret polar codes as low-density parity-check-like (LDPC-like) codes, this paper develops novel belief propagation (BP)-based decoders utilizing the concept of schedule diversity. By analyzing the impact of hidden variable nodes (VNHs), scheduling algorithms that guarantee fast convergence speeds and superior error performance are devised for the LDPC-like sparse factor graph (SPF), incorporating both standard sequential scheduling (SSS) and informed dynamic scheduling (IDS) algorithms. To further enhance error performance, cyclic redundancy check (CRC) bits are integrated into the LDPC-like SPF, and random update sequences are introduced to reduce computational complexity while implementing schedule diversity. Inspired by belief propagation list decoding, the proposed approaches eventually update messages in the LDPC-like SPF according to various schedules, thereby producing diverse decoding outcomes. Compared to the CRC-aided successive cancellation list (CA-SCL) decoder and benchmark soft-output decoders in the literature, the proposed schemes achieve comparable or improved error performance while reducing computational complexity in the high SNR region. These results serve as preliminary findings on the BP algorithm over LDPC-like SFGs and provide useful insights for the design and practical improvement of polar decoders.

Index Terms—Polar codes, low-density parity-check-like codes, belief propagation, schedule diversity, standard sequential scheduling, informed dynamic scheduling, list decoding.

I. INTRODUCTION

OVER the past decades, capacity-approaching channel codes have emerged, including the classic turbo codes [1], which are employed in the 3G UMTS [2] and 4G LTE [3] mobile broadband standards, as well as low-density parity-check (LDPC) codes [4], adopted in standards such as WiMax

[5] and DVB-S2 [6]. Both turbo and LDPC codes iteratively process the soft log-likelihood ratio (LLR) values obtained from the received signal until valid codewords are successfully decoded.

Recently, polar codes [7] have been proven to achieve channel capacity with infinite codeword lengths under successive cancellation (SC) decoding. However, when applied to finite-length codes, polar codes exhibit significantly degraded error performance. Since the recent adoption of polar codes by 3GPP as the standard codes for 5G enhanced mobile broadband (eMBB) uplink control channels [8], research on polar decoding algorithms has attracted significant attention. Due to their relatively low computational complexity, SC-based decoding algorithms remain the preferred solution. The SC list (SCL) decoder [9] was proposed to successfully achieve the maximum likelihood (ML) bound of polar codes in high-SNR regions. Furthermore, by concatenating a cyclic redundancy check (CRC) code with a polar code, the code structure is modified, enabling bit error rate (BER) performance superior to that of LDPC codes when using the CRC-aided SCL (CA-SCL) decoder [9], [10], [11].

The original SC decoder presented in [7] was not designed to support soft-input and soft-output (SISO) information. A notable exploration of advanced SC-based soft-output polar decoders is the soft list decoder [12], which produces the LLR values using the survivor candidates in SCL decoding. In contrast, the soft cancellation (SCAN) decoder was proposed in [13] without employing the concept of list decoding. Although its message-passing process follows that of SC decoding, SCAN generates soft outputs with significantly lower complexity in both storage and computation while also providing improved error performance compared to conventional SC decoding. Recently, a fast soft-output list (FastSOL) decoder, which integrates advantages from both SCL and SCAN decoders, was proposed in [14]. It achieves error performance comparable to that of the CA-SCL decoder while significantly reducing computational complexity.

A major drawback of SC-based decoding lies in its inherently serial nature, which leads to long decoding latency and, consequently, reduced throughput. In contrast, belief propagation (BP)-based decoders operate without such constraints, making them attractive candidates for high-data-rate applications, particularly from an implementation perspective. The original Arıkan BP decoder for polar codes operates on the polar encoding factor graph [15]. As noted in [16], the parity-check matrix derived from the polar encoding factor

Received 5 March 2025; revised 5 September 2025 and 2 December 2025; accepted 9 December 2025. Date of publication 15 December 2025; date of current version 2 January 2026. This work was supported in part by the National Science and Technology Council, Taiwan, under Grant NSTC 114-2218-E-110-005 and Grant NSTC 114-2221-E-110-033, and in part by the Sixth Generation Communication and Sensing Research Center funded by the Higher Education SPROUT Project, the Ministry of Education of Taiwan. The associate editor coordinating the review of this article and approving it for publication was R. Asvadi. (*Corresponding author: Yen-Ming Chen.*)

Cheng-Jun You, Yi-Hsin Chen, and Chen-Wei Hu were with the Institute of Communications Engineering, National Sun Yat-sen University, Kaohsiung 80424, Taiwan (e-mail: youchengjun0@gmail.com; a0970640929@gmail.com; dreamstar31642@gmail.com).

Huang-Chang Lee is with the Department of Electrical Engineering, National Taiwan Ocean University, Keelung 202301, Taiwan (e-mail: hclee@mail.ntou.edu.tw).

Yen-Ming Chen is with the Institute of Communications Engineering, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan (e-mail: emerychen.cm@gmail.com).

Digital Object Identifier 10.1109/TCOMM.2025.3644495

0090-6778 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: Yonsei Univ. Downloaded on June 24, 2026 at 05:49:30 UTC from IEEE Xplore. Restrictions apply.

graph is high-density and contains numerous short cycles. As a result, applying BP decoding to the factor graph based on the parity-check matrix of a polar code can easily result in decoding failures. Recently, Cammerer et al. introduced pruning techniques [17] capable of transforming the polar encoding factor graph into an LDPC-like sparse factor graph (SFG). This method involves pruning redundant variable nodes (VNs) and check nodes (CNs) to obtain an SFG suitable for iterative processing. As a result, the sum-product algorithm (SPA) decoder can be used to decode polar codes with nearly identical BER performance compared to the original Arkan BP decoder.

An effective method to improve the error performance of BP-based decoding approaches is to increase the number of candidate codewords during decoding. It has been observed that the stages of the polar encoding graph of length N can be permuted, resulting in $(\log_2 N)!$ different graphs that exhibit identical encoding behavior. Based on this observation, the BP list (BPL) decoding algorithm [18] has been proposed, utilizing the concept of decoding with parallel BP decoders. The BPL decoder comprises multiple parallel, independent BP decoders, each operating on a different permutation of the polar code's encoding graph. A list of decoded codewords is then generated, and the codeword vector closest to the received signal vector in terms of Euclidean distance is ultimately selected. However, due to the parallel message propagation and the correlations introduced by loops within the graph, it is challenging for the BP algorithm to accurately estimate the probability of a candidate codeword being correct. Consequently, BP-based decoding generally does not achieve the same level of performance as SC-based decoding.

On the other hand, the conventional decoding process of LDPC codes is based on BP decoding, which considers various message-passing schedules. Flooding, also referred to as simultaneous scheduling, is the most commonly used scheduling approach. In each iteration, Flooding decoding updates all VNs simultaneously, followed by the simultaneous update of all CNs. In contrast, the non-simultaneous decoding can be categorized into two types: standard sequential scheduling (SSS) [19], [20] and informed dynamic scheduling (IDS) [21], [22], [23], [24]. Layered belief propagation (LBP) decoding, which follows a predetermined order of message updates [19], is a typical SSS strategy. It has been demonstrated in [19] that LBP achieves a convergence speed twice as fast as Flooding decoding, without incurring additional computational complexity. However, sequential updating presents the challenge of determining the optimal order of message updates to achieve the best convergence speed and/or error performance. In fact, the current state of messages in the graph can be leveraged to dynamically adjust the schedule, leading to research in IDS.

Considering dynamic schedules, various IDS approaches have been investigated in the literature. Residual belief propagation (RBP) decoding [21] utilizes the residual value of message updates, which is defined as the difference between message values before and after an edge update. This residual value serves as a reference for determining the order of message updates. The intuition behind RBP is that not all messages contribute equally to convergence. Sending a message that

significantly differs from its previous value is likely to be more informative and can accelerate the transfer of information [21]. Simulation results indicate that RBP achieves faster convergence compared to SSS approaches. However, the error performance deteriorates after convergence when the number of iterations becomes sufficiently large. This phenomenon is a common characteristic of greedy algorithms, which are designed to rapidly identify solutions but often face challenges in converging to the optimal one [22].

To mitigate the greedy behavior in RBP, node-wise residual belief propagation (NWRBP) decoding was proposed in [22], where all check-to-variable (C2V) messages corresponding to the selected CN are updated simultaneously. Although the convergence speed of NWRBP is generally slower than that of RBP for a small number of iterations, NWRBP consistently outperforms SSS strategies, such as LBP, across all iteration counts. Similarly, variable-to-check residual belief propagation (VCRBP) decoding, reported in [23], is designed based on variable-to-check (V2C) residual values. It has been observed that VCRBP demonstrates superior convergence speed and error performance compared to both RBP and NWRBP [23].

Inspired by the BPL decoder, performing BP decoding on various permuted polar encoding factor graphs is analogous to BP decoding on different pruned LDPC-like SFGs. At a deeper level, the results achieved by BPL decoding suggest that messages associated with the VNs in the encoding graph can be updated according to different schedules, ultimately yielding similar or enhanced error performance. This insight has led us to explore possible combinations of scheduling methods and list decoding approaches based on the LDPC-like SFG. The contributions of this paper are summarized as follows.

- 1) When directly applying various LDPC scheduling strategies reported in the literature to the LDPC-like SFG, the error performance trends differ significantly from those observed in conventional LDPC decoding. These phenomena have been analyzed, leading to the development of scheduling strategies specifically tailored for LDPC-like SFGs.
- 2) Two scheduling strategies, referred to as the LBP with partially-determined update sequence (LBP-PUS) and Shuffle LBP-PUS, are designed based on the advantages of SSS and IDS decoding strategies. Both strategies achieve faster convergence and improved error performance compared to benchmark scheduling strategies.
- 3) By leveraging systematic polar encoding, CRC check equations are integrated with the LDPC-like SFG. As a result, the CRC CNs are incorporated into the iterative decoding process, enhancing the converged error performance and enabling early termination using both the LDPC-like SFG and the CRC bits.
- 4) By introducing random update sequences, computational complexity can be reduced, and the concept of schedule diversity [20] can be implemented with the aid of bit-flipping techniques [25], thereby significantly improving error performance.
- 5) Compared to the benchmark decoders BPL and SCAN, the proposed schemes without schedule diversity exhibit

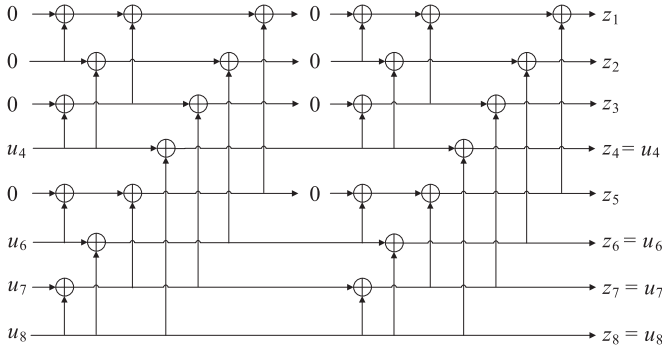


Fig. 1. An example of the systematic polar encoding for an $(8, 4)$ polar code.

superior error performance. Moreover, when evaluated against the benchmark decoders Soft List, CA-SCL, and FastSOL, the proposed schemes with schedule diversity achieve comparable or enhanced error performance while reducing computational complexity in the high SNR region. These results also serve as preliminary findings on the BP algorithm in LDPC-like SFG, and reveal promising potential directions for future research.

The remainder of this paper is organized as follows. Section II introduces the fundamentals of polar encoding, LDPC-like SFGs, and BP decoding using IDS strategies. To progressively present the development of the final algorithms, the first stage, corresponding to Sections III-A through III-C, considers the case where CRC bits are excluded from the decoding process, during which scheduling algorithms that provide both rapid convergence and excellent converged error performance are constructed. The second stage, described in Sections III-D and III-E, investigates scheduling algorithms that incorporate CRC bits into the decoding procedures and adopt a random update sequence, thereby facilitating the realization of schedule diversity. The proposed decoding algorithms that integrate both schedule diversity and the bit-flipping strategy are detailed in Section IV. Finally, Section V concludes the paper.

II. PRELIMINARIES

A. Polar Encoding

Polar code is an (N, K) linear block code where $N = 2^n$ for some n values. Considering a length- N binary vector $u_1^N \equiv [u_1, u_2, \dots, u_N]$ which includes K information bits and $N - K$ frozen bits, a length- N polar codeword vector $x_1^N \equiv [x_1, x_2, \dots, x_N]$ can be generated as

$$x_1^N = u_1^N \mathbf{G}_N, \quad (1)$$

where $\mathbf{G}_N \equiv \mathbf{B}_N \mathbf{F}_2^{\otimes n}$ denotes the $N \times N$ generator matrix, $\mathbf{B}_N$ denotes the bit-reversal matrix, $\mathbf{F}_2 \equiv \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\otimes$ denotes the Kronecker product. Instead of conventional polar encoding, the systematic polar encoding process is adopted throughout this paper to enhance error performance and to facilitate the design of the proposed algorithm. A systematic polar codeword vector z_1^N can be generated as

$$z_1^N = u_1^N \mathbf{G}_N \mathbf{E}_N \mathbf{G}_N, \quad (2)$$

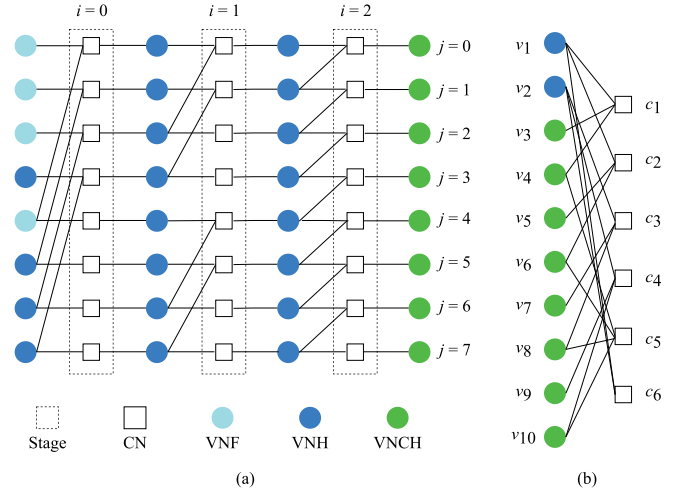


Fig. 2. An example of the graph representations for an $(8, 4)$ polar code. (a) The polar encoding factor graph; (b) The corresponding LDPC-like SFG.

where $\mathbf{E}_N$ is defined as the $N \times N$ unfrozen bit-reversal location matrix. An example is depicted in Fig. 1, where

$$\mathbf{E}_N = \begin{bmatrix} 0, 0, 0, 0, 0, 0, 0, 0 \\ 0, 0, 0, 0, 0, 0, 0, 0 \\ 0, 0, 0, 0, 0, 0, 0, 0 \\ 0, 0, 0, 1, 0, 0, 0, 0 \\ 0, 0, 0, 0, 0, 0, 0, 0 \\ 0, 0, 0, 0, 0, 1, 0, 0 \\ 0, 0, 0, 0, 0, 0, 1, 0 \\ 0, 0, 0, 0, 0, 0, 0, 1 \end{bmatrix} \quad (3)$$

re-assign the frozen values as zero after the first half of the encoding process. For details on systematic polar encoding, please refer to [26].

B. LDPC-Like Sparse Factor Graph

In [16], applying BP decoding to the factor graph derived from the parity-check matrix $\mathbf{H}$ of a polar code may lead to decoding failures due to the high density of the graph. To tackle this issue, several pruning techniques, which are directly applied to the encoding factor graph of the polar code, are proposed in [17]. In Fig. 2, we first define the channel variable nodes (VNCH) as the VNs corresponding to the transmitted polar code bits that can acquire the associated channel LLR values. We further define the hidden variable nodes (VNH) as the intermediate VNs remaining after pruning the middle part of the polar encoding graph. Consequently, the VNHs lack any information, meaning their corresponding channel LLR values should be set to zero. Lastly, the frozen variable nodes (VNF) are defined as the VNs that correspond to the frozen bits, where the *a priori* LLR values should be set to $+\infty$ if the frozen bit values are set to zero.

In [17], six pruning scenarios are presented, allowing frozen nodes, degree-1 VNCHs and degree-2 CNs, degree-2 VNHs, and several other nodes to be removed from the factor graph. Through these pruning operations, the encoding factor graph can ultimately be simplified into an LDPC-like SFG [17], as illustrated in Fig. 2b. Since pruning techniques are not the

main focus of this paper, readers are referred to [17] for detailed discussions of the LDPC-like SFG and the associated pruning methods. Benefiting from the low-density property of the LDPC-like SFG, iterative decoding based on the SPA can be applied. As indicated in [17], the “LDPC-like BP” incurs negligible performance loss for short-to-intermediate code lengths compared to the original Arkan BP. However, both BP decoding methods exhibit error performance significantly inferior to that of CA-SCL decoding.

C. Belief Propagation Decoding on LDPC-Like Sparse Factor Graph

We consider a pruned LDPC-like SFG containing N' VNs and M CNs, where $N' > N$. Let v_j denote the j -th VN and c_i denote the i -th CN in the graph, respectively. Without loss of generality, we assume that the last N nodes among the N' VNs are VNCHs, while the remaining $N' - N$ nodes are VNHs. Consequently, the channel LLR value L_j of a VN v_j can be expressed as

$$L_j = \begin{cases} 0, & 1 \leq j \leq N' - N, \\ \ln \frac{p(y_n | z_n = 0)}{p(y_n | z_n = 1)}, & N' - N + 1 \leq j \leq N', n = j \\ -N' + N, & \end{cases} \quad (4)$$

where $y_n \equiv (-1)^{z_n} + w_n$, $n = 1, 2, \dots, N$, denotes the received signal considering an AWGN channel and w_n denotes the $CN(0, 2\sigma^2)$ distributed noise signal.

When considering the BP decoding, the V2C LLR value transmitted from v_j to c_i is denoted as $m_{v_j \rightarrow c_i}$, and is given by

$$m_{v_j \rightarrow c_i} = \sum_{c_k \in N(v_j) \setminus c_i} m_{c_k \rightarrow v_j} + L_j, \quad (5)$$

where $N(v_j) \setminus c_i$ denotes the set which includes the neighboring CNs of v_j except for c_i . In contrast, the C2V LLR value transmitted from c_i to v_j is denoted as $m_{c_i \rightarrow v_j}$, and is given by

$$m_{c_i \rightarrow v_j} = \prod_{v_k \in N(c_i) \setminus v_j} \text{sign}(m_{v_k \rightarrow c_i}) \cdot \phi \left(\sum_{v_k \in N(c_i) \setminus v_j} \phi(|m_{v_k \rightarrow c_i}|) \right), \quad (6)$$

where $N(c_i) \setminus v_j$ denotes the set which includes the neighboring VNs of c_i except for v_j , $|\cdot|$ denotes the absolute value, and $\phi(x) = \phi^{-1}(x) \equiv -\log(\tanh(\frac{x}{2}))$. The two types of LLR values are iteratively updated following a specific scheduling approach until the stopping criterion is fulfilled. In this paper, the stopping criterion is defined as the triggering of certain early termination conditions, or the maximum number of iterations, I_{max} , being reached. Lastly, the hard decision is performed on the VNCHs that correspond to the information bits by using the summation of all incoming LLR values, which is defined as

$$L(v_j) \equiv \sum_{c_k \in N(v_j)} m_{c_k \rightarrow v_j} + L_j. \quad (7)$$

D. Node-Wise Residual Belief Propagation

In conventional LDPC decoding, the current state of messages in the factor graph, reflected by their residual values, can be used to dynamically refine the decoding schedule. The resulting IDS strategies incorporate the well-known RBP algorithm [21] as one of their key components. RBP updates C2V messages on an edge basis: all edges are sorted according to their residual values, and only one edge is updated at a time. As a result, RBP tends to excessively update messages on less reliable edges, leading to greedy behavior. In contrast, NWRBP [22] updates and propagates all C2V messages corresponding to the selected CN simultaneously, rather than propagating only the message on a specific edge, thereby mitigating its greedy behavior.

NWRBP involves a sequence of check-node updates. Different from the LBP which adopts a pre-determined order, NWRBP dynamically selects the next CN to be updated based on the C2V residual values, which is defined as

$$r(m) = |f(m) - m|, \quad (8)$$

where $f(m)$ denotes the (estimated) message after the update, while m represents the message before the update. Consequently, the CN c_i with the largest $r(m_{c_i \rightarrow v_j})$ value is selected. Then, $m_{c_i \rightarrow v_k}$ is updated for all $v_k \in N(c_i)$, and $r(m_{c_i \rightarrow v_k})$ is set to zero for all $v_k \in N(c_i)$. Next, $m_{v_k \rightarrow c_a}$ is generated and propagated for all $c_a \in N(v_k) \setminus c_i$. Finally, the corresponding residual values $r(m_{c_a \rightarrow v_b})$ are calculated for all $v_b \in N(c_a) \setminus v_k$. This process repeats until the stopping criterion is satisfied.

E. Variable-to-Check Residual Belief Propagation

VCRBP [23] is an alternative algorithm designed to mitigate the greedy behavior of RBP. The main difference between VCRBP and RBP lies in how the residual values are computed. VCRBP calculates the residual value by measuring the difference in V2C messages before and after the update. Consequently, it updates the C2V messages of the CN connected to the selected edge with the largest V2C residual value.

In the first step, the largest $r(m_{v_j \rightarrow c_i})$ value is identified and set to zero. Then, $m_{c_i \rightarrow v_a}$ is updated for all $v_a \in N(c_i) \setminus v_j$. Finally, $m_{v_a \rightarrow c_b}$ is updated for all $c_b \in N(v_a) \setminus c_i$, and the correspondent residual values $r(m_{v_a \rightarrow c_b})$ are calculated. This process repeats until the stopping criterion is satisfied.

III. PROPOSED SCHEDULING ALGORITHMS FOR POLAR CODES BASED ON LDPC-LIKE SPARSE FACTOR GRAPHS

In [18], the BPL decoding algorithm is proposed, wherein the stages of the polar encoding factor graph can be permuted to generate multiple graphs that maintain identical encoding behavior, thereby enabling the use of parallel BP decoders. After the decoding process, polar codewords are re-encoded using these graphs and their corresponding decoded information bits. The information bits corresponding to the re-encoded codeword that guarantees the minimum Euclidean distance when compared to the received signal vector are chosen as the final decoded information bits. Consequently, the list decoding feature of this approach improves the error performance of

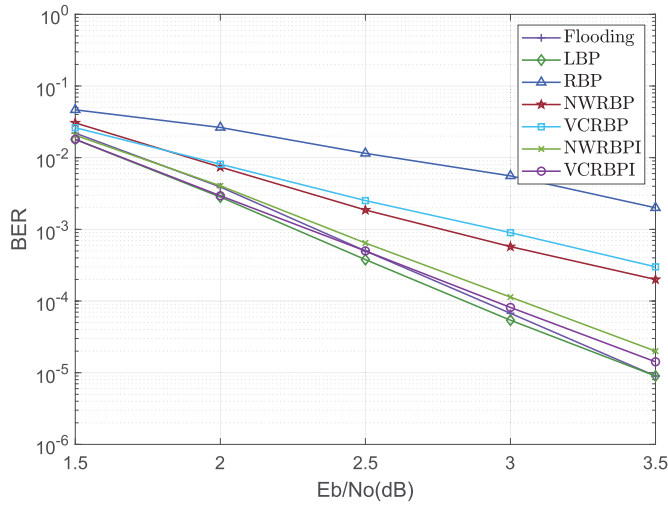


Fig. 3. Comparison of BER performance among the Flooding decoding, SSS decoding, and IDS decoding strategies on the LDPC-like SFG. $I_{\max} = 100$.

polar decoding. The results reported in [18] actually imply that messages related to the VNs existed in the encoding graph can be updated according to different schedules, while ultimately achieving similar or improved error performances. Inspired by [18], scheduling strategies tailored for LDPC-like SFGs are developed in this section, aiming to improve error performance and increase convergence speed.

Following the approach reported in [17], the Bhattacharyya parameter method [7] is utilized to construct a polar code with dimensions $(N, K) = (1024, 488 + 24)$ and a code rate of $R = 488/1024$, where 24 denotes the length of the CRC bits. The considered CRC generator polynomial is given by $g(x) = x^{24} + x^{23} + x^6 + x^5 + x + 1$. In Sections III-A to III-C, the early termination is triggered if the CRC check is satisfied. When the polar encoding graph is directly converted into an LDPC-like factor graph, the resulting graph has $N' = 11264$ and $M = 10240$. With pruning techniques applied, these dimensions are significantly reduced to $N' = 2335$ and $M = 1823$.

A. Experiments on Applying Conventional Scheduling Strategies to the LDPC-Like Sparse Factor Graph

In this subsection, LBP [19] is adopted as the benchmark strategy for SSS decoding, while RBP [21], NWRBP [22], and VCRBP [23] are adopted as the benchmark strategies for IDS decoding. When applying these strategies to the LDPC decoding, several trends have been reported in the literature [24]. First, RBP exhibits extremely fast convergence speed, but its converged error performance is unsatisfactory. Second, although LBP and NWRBP provide slower convergence speeds compared to RBP, they outperform Flooding decoding in both convergence speed and converged error performance. Third, the node-wise IDS strategies, i.e., NWRBP and VCRBP, outperform LBP in both convergence speed and converged error performance.

When these strategies are directly applied to the LDPC-like SFG, the corresponding BER and convergence performances are shown in Figs. 3 and 4, respectively. It is observed that the RBP and LBP strategies follow similar trends to those seen

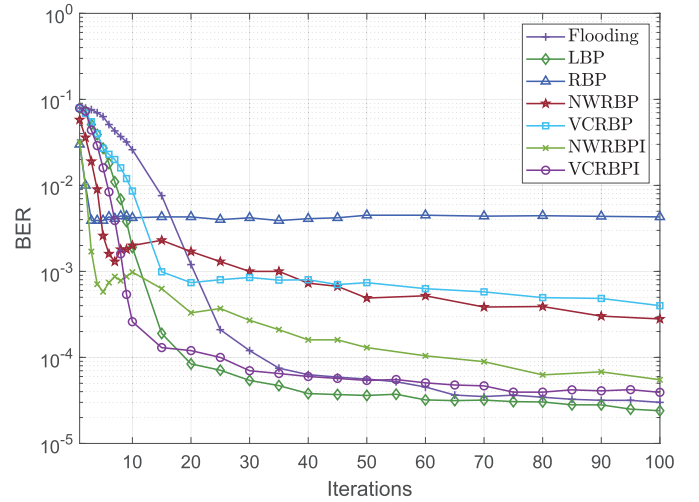


Fig. 4. Comparison of convergence performance among the Flooding decoding, SSS decoding, and IDS decoding strategies on the LDPC-like SFG. $E_b/N_0 = 3.214$ dB.

in LDPC decoding. However, compared to LBP and Flooding, the node-wise IDS strategies exhibit poorer converged error performance when applied to the LDPC-like SFG.

Building on this intriguing phenomenon, we further observe that the pruned SFG contains no 4-cycles but a substantial number of 6-cycles. Moreover, CNs involved in substantially more 6-cycles tend to exhibit excessive update counts, leading to greedy behaviors. To validate this assumption, we impose a simple constraint on the node-wise IDS decoding strategies: each CN is allowed to be updated only once per iteration, thereby ensuring a fair distribution of update resources. The resulting algorithms are referred to as NWRBPI and VCRBPI.

In Figs. 3 and 4, it can be observed that both the convergence speed and the converged error performance of NWRBPI and VCRBPI are superior to those of the original NWRBP and VCRBP algorithms, respectively. As shown in Fig. 4, NWRBPI and VCRBPI generally guarantee faster convergence speeds compared to NWRBP, VCRBP and LBP. This suggests that by limiting the number of node updates, we can not only address the issue of greedy behavior but also reallocate computing resources more efficiently, thereby accelerating the convergence speed. However, as shown in Fig. 3 the BER performances of NWRBPI and VCRBPI are still inferior to those of LBP and Flooding. We believe this indicates that there are still fundamental issues stemming from the structure of the LDPC-like SFG, which continue to limit its performance. In this paper, rather than modifying the SFG structures, we place our emphasis on receiver design and address the issue of greediness by investigating appropriate decoding algorithms.

If we further examine the SFG, a significant number of VNHs that do not receive channel LLR values can be identified. In Section III-B, we analyze the impact of the VNHs on BP decoding. Accordingly, two decoding algorithms are proposed in Section III-C, based on the integration of SSS and IDS decoding strategies. Subsequently, in Section III-D, we discuss the modifications to the proposed algorithms based on the integration of CRC CNs. Lastly, in Section III-E, the impact of using random updates is analyzed.

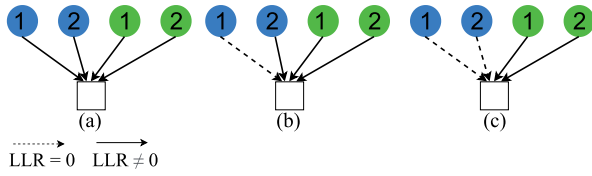


Fig. 5. Three possible cases of the message passing for a degree-4 CN that connects to 2 VNHs and 2 VNCHs.

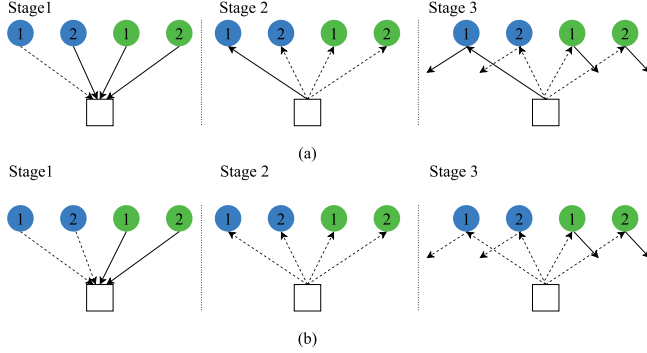


Fig. 6. Three stages of the message passing for the case given in (a) Fig. 5b and (b) Fig. 5c.

B. Propagation of Zero Values

In Fig. 5, a sub-graph containing one CN and its neighboring two VNHs and two VNCHs is considered, illustrating three possible cases of message passing. In Fig. 5a, during the previous update, both VNHs may have received nonzero C2V values from other CNs, allowing them to transmit nonzero V2C values to this CN in the current update, even if they do not possess any channel LLR values. In contrast, in the second case shown in Fig. 5b, VNH 1 did not receive any nonzero C2V value in the previous update. As a result, it transmits a zero V2C value to the CN in the current update. The three stages of the message passing for this case are illustrated in Fig. 6a. It is shown that although the CN generates and transmits a zero C2V value to VNH 2 according to (6), the messages from VNCHs 1 and 2 are still successfully propagated to VNH 1 and subsequently to other CNs connected to VNH 1 in later updates. Finally, the three stages of the message passing for the third case, depicted in Fig. 5c, are illustrated in Fig. 6c. It is shown that if both VNHs transmit zero V2C values to the CN, the messages from VNCHs 1 and 2 cannot be successfully propagated to any of the VNHs. From this example, it can be deduced that if there are a sufficient number of CNs connected to more than one VNH, zero values may continue to propagate during the iteration process. A closer examination of the C2V messages exchanged in the SFG reveals that a significant number of these values are influenced by the zero values, resulting in relatively small amplitudes that increase slowly throughout the iteration process. Consequently, the IDS strategy is less likely to select these nodes during the decoding process, thereby hindering soft-message exchange in the SFG. Since VNHs can be interpreted as punctured nodes within the SFG, we attribute this phenomenon to the

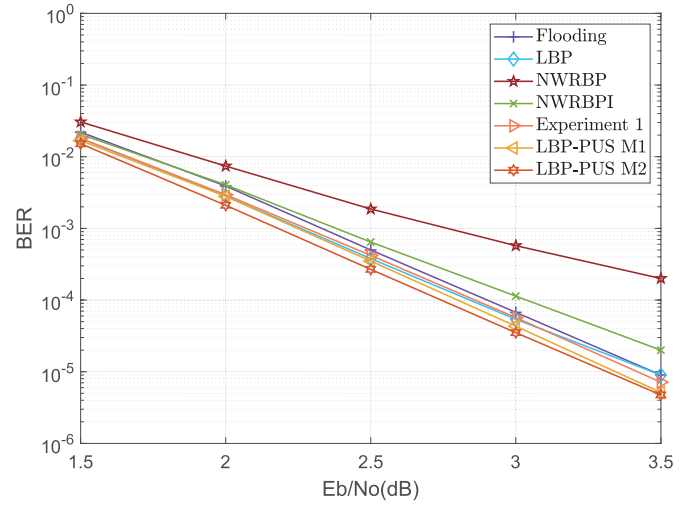


Fig. 7. Comparison of BER performance between various benchmark methods and the proposed methods on the LDPC-like SFG. $I_{\max} = 100$.

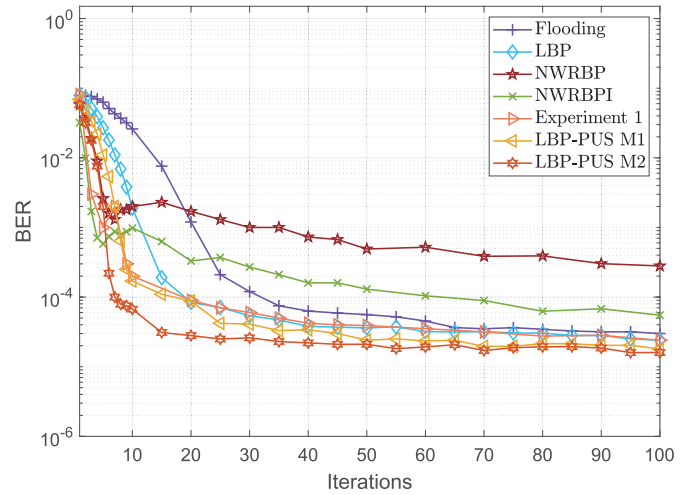


Fig. 8. Comparison of convergence performance between various benchmark methods and the proposed methods on the LDPC-like SFG. $E_b/N_0 = 3.214$ dB.

fact that the SFG involves a substantially larger number of punctured nodes than the punctured LDPC codes discussed in the literature, thereby prevents effective recovery during the iteration process under conventional IDS-based algorithms.

Consequently, an experiment is conducted based on this deduction:

Experiment 1: Initially, the NWRBPI algorithm is applied to a set of CNs, excluding those connected to more than one VNH. After each CN in this set is updated, it can be observed that some of the previously excluded CNs now have part of their connected VNHs receiving non-zero C2V values from the previous update. These VNHs are now able to broadcast non-zero V2C values in the subsequent decoding process. Consequently, if any of the previously excluded CNs is now connected to only one VN that broadcasts zero V2C values, the CN should be re-included in the CN set. The NWRBPI algorithm is then applied again to the CN set. The decoding process terminates once the stopping criterion is satisfied.

As demonstrated in Figs. 7 and 8, this experiment shows a significant improvement in converged error performance

compared to NWRBPI. Additionally, when compared to LBP, the experiment exhibits comparable converged error performance but with a much faster convergence speed.

C. Proposed Scheduling Algorithm

The proposed scheduling algorithm is designed based on the experimental results provided in Section III-B. As shown in Fig. 6b, if a CN receives two or more zero-valued LLRs, all of the LLR values generated by this CN and forwarded to the connected VNs will be zero. Consequently, it can be confirmed that the scenario in which a CN broadcasts zero values is effectively equivalent to the scenario where all C2V residual values associated with this CN are zero.

In contrast, during the final stages of the decoding process, if an LLR value transmitted along an edge has converged, the corresponding C2V residual value also approaches zero. To accelerate the decoding process, the CNs that have already delivered converged LLR values should also be excluded from the set of CNs to be updated.

Algorithm 1 The LBP-PUS Decoding Algorithm

```

1: Initialize  $m_{c_i \rightarrow v_j}$  values as 0,  $\forall i, j$ .
2: Initialize  $m_{v_j \rightarrow c_i}$  values as  $L_j$ ,  $\forall i, j$ .
3: Compute  $r(m_{c_i \rightarrow v_j})$  values,  $\forall i, j$ .
4: Compute the maximum C2V residual value  $r_{c_i, \max} \equiv \max \{r(m_{c_i \rightarrow v_j}) \mid v_j \in N(c_i)\}$  and calculate the threshold  $\mu$  based on  $r_{c_i, \max}$ .
5: If any  $r_{c_i, \max}$  value is larger than  $\mu$ ,  $c_i$  is included in the set  $\mathcal{Q}$ . Sort  $\mathcal{Q}$  in an descending order based on  $r_{c_i, \max}$ .
6: for every  $c_i \in \mathcal{Q}$  do
7:   for every  $v_k \in N(c_i)$  do
8:     Generate and propagate  $m_{c_i \rightarrow v_k}$ 
9:   for every  $c_a \in N(v_k) \setminus c_i$  do
10:    Generate and propagate  $m_{v_k \rightarrow c_a}$ 
11:   end for
12: end for
13: end for
14: if the stopping criterion is not satisfied, then
15:   Return to line 3.
16: end if

```

To address these requirements, and drawing on the principles of both the NWRBPI and LBP algorithms, the proposed approach is summarized in Algorithm 1, where the parameter μ is currently set to zero. Unlike RBP-based algorithms, the update sequence here is predetermined and remains fixed until all CN updates in the sequence are completed. This strategy is more consistent with the conventional LBP algorithm. Accordingly, we refer to this method as LBP with partially-determined update sequence (LBP-PUS).

In Steps 3 and 4 of Algorithm 1, the residual values for all edges connected to a specific CN are calculated, and the maximum C2V residual value is taken as the representative residual value, denoted as $r_{c_i, \max}$, for that CN. In Step 5, CNs with $r_{c_i, \max}$ values larger than the threshold μ are included in $\mathcal{Q}$, which is defined as the set of CNs to be updated. The CNs in $\mathcal{Q}$ are then sorted in descending order based on their $r_{c_i, \max}$ values. Finally, the CNs in $\mathcal{Q}$ are updated using the LBP

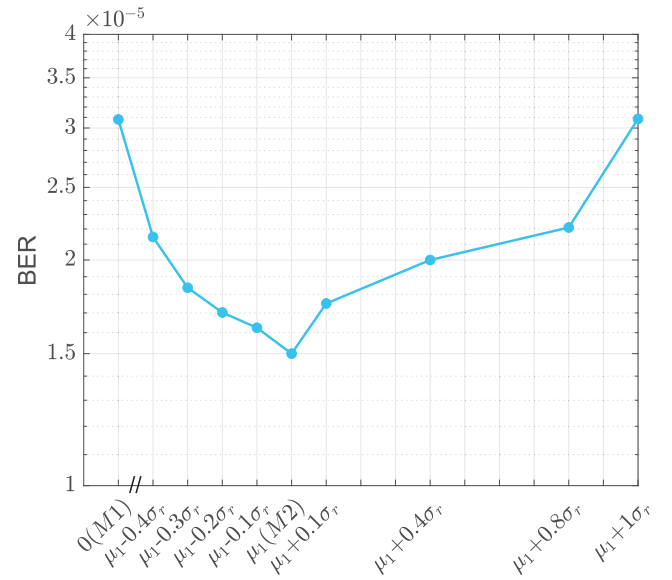


Fig. 9. BER versus various μ values when $E_b/N_0 = 3$ dB.

algorithm. These steps are repeated until the stopping criterion is satisfied. For more details, please refer to Algorithm 1.

During the early stages of the decoding process, both zero LLR values and non-zero LLR values with small magnitudes can significantly influence the convergence behavior. Furthermore, in the final stages of decoding, CNs nearly converged can be excluded from the set $\mathcal{Q}$ in advance. As a result, the value of μ can be appropriately adjusted to be greater than zero. We refer to the case $\mu = 0$ as LBP-PUS Method 1 (LBP-PUS M1), whereas the case $\mu > 0$ is referred to as LBP-PUS Method 2 (LBP-PUS M2).

Focusing on LBP-PUS M2, the BER performance under various μ values at $\frac{E_b}{N_0} = 3$ dB is illustrated in Fig. 9. Let σ_r denote the standard deviation of the non-zero $r_{c_i, \max}$ values. As shown in Fig. 9, any deviation of the threshold from μ_1 (the mean of the non-zero $r_{c_i, \max}$ values across CNs), either upward or downward, degrades the BER performance. Therefore, we set $\mu = \mu_1$ in this paper. As shown in Figs. 7 and 8, relative to the experimental scheme, LBP-PUS M1 achieves a better converged error rate, and LBP-PUS M2 further improves both the convergence speed and the converged error performance.

Given that RBP and VCRBP are both fundamental algorithms in IDS research, building on similar considerations, we further adopt both the VCRBPI and LBP algorithms and develop a new algorithm named Shuffled LBP-PUS. In this algorithm, similarly, μ is calculated based on the $r_{c_i, \max}$ values, and the set $\mathcal{Q}$ is built accordingly. However, the CNs contained in $\mathcal{Q}$ are sorted based on the $\tilde{r}_{c_i, \max}$ values, which are defined as the maximum V2C residual values of a CN. Lastly, the CNs contained in $\mathcal{Q}$ are updated based on the LBP algorithm. These steps are repeated until the stopping criterion is satisfied. For more details, please refer to Algorithm 2.

In Figs. 10 and 11, it can be observed that both the LBP-PUS M2 and Shuffled LBP-PUS M2 algorithms guarantee enhanced convergence speeds and converged error performance when compared to the benchmark schemes. In

Algorithm 2 The Shuffled LBP-PUS Decoding Algorithm

- 1: Initialize $m_{c_i \rightarrow v_j}$ values as 0, $\forall i, j$.
- 2: Initialize $m_{v_j \rightarrow c_i}$ values as L_j , $\forall i, j$.
- 3: Compute $r(m_{c_i \rightarrow v_j})$ and $r(m_{v_j \rightarrow c_i})$, $\forall i, j$.
- 4: Compute the **maximum V2C residual value** $\tilde{r}_{c_i, \max} \equiv \max \{r(m_{v_k \rightarrow c_i} \mid v_k \in N(c_i))\}$ and define $v_{c_i, \max}$ as the VN in $N(c_i)$ that provides the $\tilde{r}_{c_i, \max}$ value.
- 5: Compute the **maximum C2V residual value** $r_{c_i, \max} \equiv \max \{r(m_{c_i \rightarrow v_j} \mid v_j \in N(c_i))\}$ and Calculate the threshold μ based on $r_{c_i, \max}$.
- 6: If any $r_{c_i, \max}$ value is larger than μ , c_i is included in the set $\mathcal{Q}$. Sort $\mathcal{Q}$ in a descending order based on $\tilde{r}_{c_i, \max}$.
- 7: **for** every $c_i \in \mathcal{Q}$ **do**
- 8: **for** every $v_k \in N(c_i) \setminus v_{c_i, \max}$ **do**
- 9: Generate and propagate $m_{c_i \rightarrow v_k}$
- 10: **for** every $c_a \in N(v_k) \setminus c_i$ **do**
- 11: Generate and propagate $m_{v_k \rightarrow c_a}$
- 12: **end for**
- 13: **end for**
- 14: **end for**
- 15: **if** the stopping criterion is not satisfied, **then**
- 16: Return to line 3.
- 17: **end if**

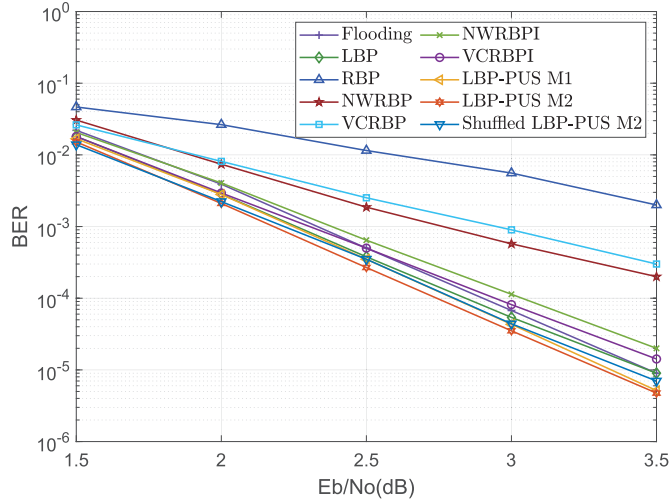


Fig. 10. Comparison of BER performance between several benchmark methods and the proposed methods on the LDPC-like SFG. $I_{\max} = 100$.

particular, LBP-PUS M2 shows a gain of approximately 0.156 dB compared to LBP and Flooding at a BER of 10^{-5} .

D. Modifications When Considering the Integrated Sparse Factor Graph

The concatenation of CRC bits alters the code structure of polar codes and significantly enhances BER performance when using the CRC-aided SCL (CA-SCL) decoder [9], [10], [11]. Consequently, it is hypothesized that the error performance of BP-based designs could be further improved if CRC bits are properly incorporated into the iterative process. On the other hand, CRC check satisfaction was used as the criterion for early termination in the previous subsections. In contrast, [27] uses the satisfaction of the parity check equations as the

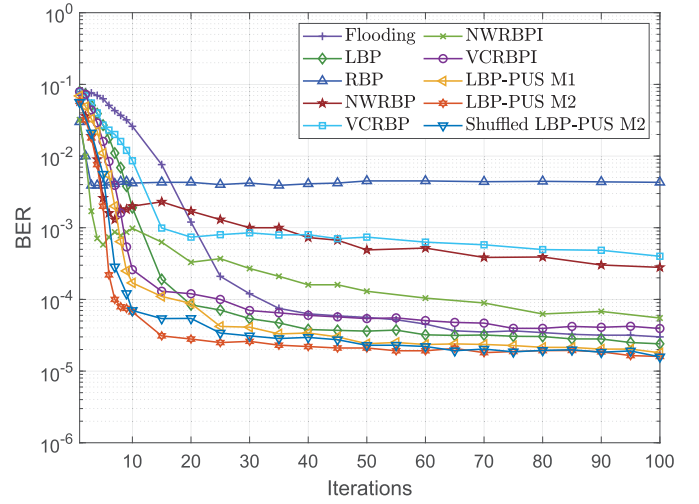


Fig. 11. Comparison of convergence performance between several benchmark methods and the proposed methods on the LDPC-like SFG. $E_b/N_0 = 3.214$ dB.

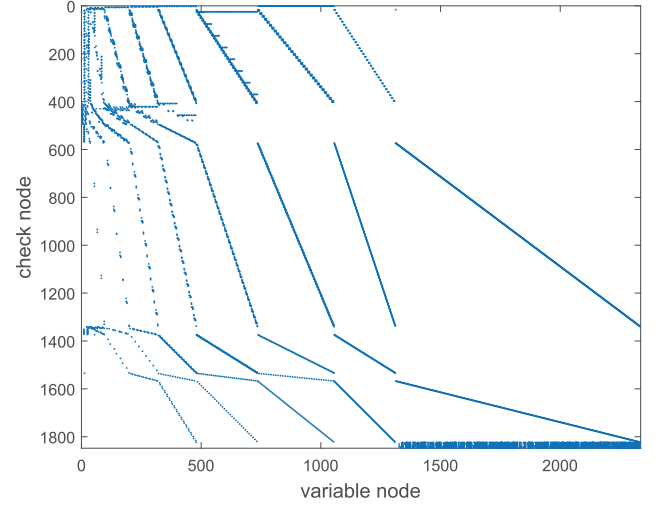


Fig. 12. Illustration of the structure for a (2335, 1847) integrated SFG considering 24 CRC CNs.

criterion for early termination. Building on both concepts, this paper leverages systematic polar encoding to integrate CRC check equations with the LDPC-like SFG. As a result, the CRC CNs actively participate in the iterative decoding process. Furthermore, by assessing the satisfaction of CNs within the integrated SFG, early termination can be determined based on both the LDPC-like SFG and the CRC bits.

We first assume that after the systematic polar encoding and graph pruning, the information bits are transmitted through the VNCHs that are distributed in the latter region of all the VNs. Consequently, these VNCHs are checked not only by the CNs within the LDPC-like SFG but also through the CRC equations. If we regard these CRC equations as additional CNs and integrate them into the SFG, a high-density subgraph would be generated in the lower right corner of the integrated SFG, as demonstrated in Fig. 12, where the blue dots represent the connections between specific CNs and VNs. It is noted that in the simulation results provided in the previous three subsections, the dimension of the LDPC-like SFG is $(N', M) = (2335, 1823)$. Beginning from this subsection, the

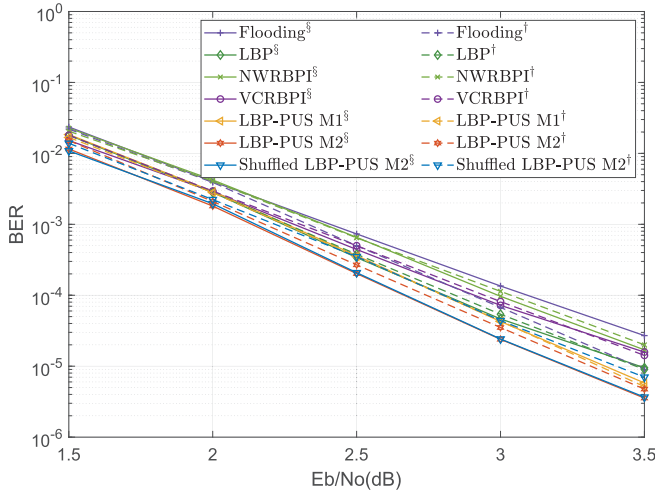


Fig. 13. Comparison of the BER performance between the schemes using the original SFG (marked with †) and the integrated SFG (marked with §). $I_{\max} = 100$.

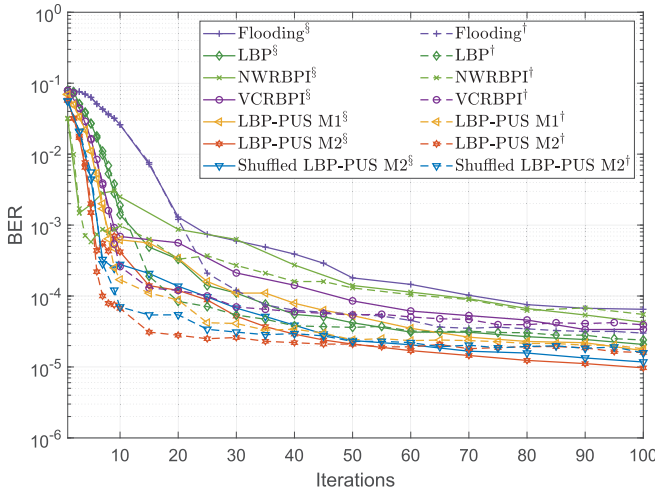


Fig. 14. Comparison of the convergence performance between the schemes using the original SFG (marked with †) and the integrated SFG (marked with §). $E_b/N_0 = 3.214$ dB.

dimension of the SFG is modified to $(N', M) = (2335, 1847)$ due to the integration of 24 additional CRC CNs.

Figs. 13 to 16 present two scenarios for comparison. The curves marked with † correspond to Scenario 1, in which the original SFG is employed during the iterative decoding process, with the satisfaction of CRC bits serving as the criterion for early termination. Conversely, the curves marked with § represent Scenario 2, where the integrated SFG is utilized in the iterative decoding process, and early termination is determined by the satisfaction of all CNs within the integrated SFG. Additionally, the second scenario demonstrates that the CRC CNs actively participate in the iterative decoding process.

The BER and convergence performances for various scheduling algorithms are reported in Figs. 13 and 14, respectively. In Fig. 13, it is shown that only certain scheduling algorithms (NWRBPI, LBP-PUS M2, and Shuffle LBP-PUS M2) guarantee improved BER performance in Scenario 2. Fig. 14 further illustrates that, although the aforementioned algorithms demonstrate improved error performance, nearly

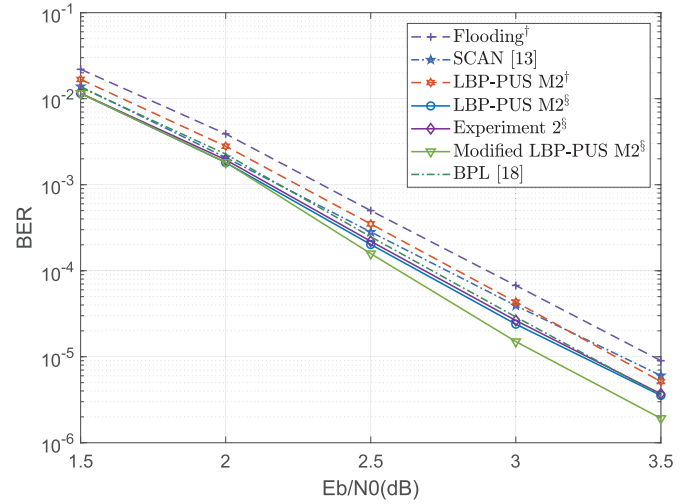


Fig. 15. Comparison of BER performance between the proposed and benchmark schemes, considering the original SFG (labeled with †) and the integrated SFG (labeled with §). $I_{\max} = 100$.

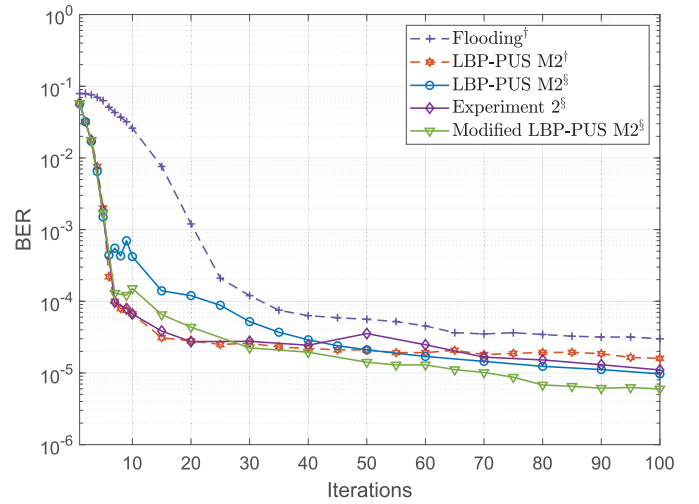


Fig. 16. Comparison of convergence performance between the proposed and benchmark schemes, considering the original SFG (labeled with †) and the integrated SFG (labeled with §). $E_b/N_0 = 3.214$ dB.

all scheduling algorithms experience slower convergence when using the integrated SFG. This suggests that incorporating the CRC CNs into the iterative decoding process may enhance the converged error performance eventually but at the cost of requiring a large number of iterations. Therefore, a modification aimed at accelerating the convergence speed is necessary.

Given that the CRC parity-check sub-matrix is a high-density matrix containing numerous short cycles, the convergence speed may decrease because the CRC CNs can be excessively updated during the early stages of iteration. To temporarily suspend the updating of CRC CNs until the soft messages generated by other CNs become relatively stable, we conducted the following experiment:

Experiment 2: Based on LBP-PUS M2, the CRC CNs are excluded from $\mathcal{Q}$ during the first κ iterations. The experimental value of κ is set to 50. As shown in Figs. 15 and 16, although the convergence speed of the experimental result is improved compared to LBP-PUS M2 using the integrated SFG, the converged error performance is degraded. This phenomenon

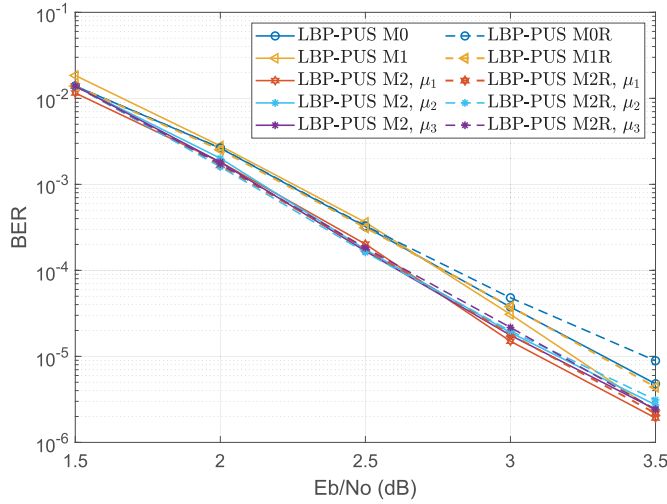


Fig. 17. An analysis of the impact of using random update sequences.

indicates that the update of certain CRC CNs remains essential during the early stages of the decoding process.

Based on these observations, the modified LBP-PUS M2 algorithm is devised, in which only the CRC CN with the highest $r_{c_i, \max}$ value is included each generation of Q . As shown in Figs. 15 and 16, the modified LBP-PUS M2 provides significant gains in BER performance of 0.245 dB and 0.375 dB compared to LBP-PUS M2 and Flooding using the original SFG, respectively, at a BER value of 10^{-5} . Additionally, the convergence speed is similar to that of the LBP-PUS M2 using the original SFG. Finally, compared to the SCAN and BPL algorithm respectively reported in [13] and [18], BER gains of 0.266 and 0.155 dB are respectively observed.

Hereafter, we consider the integrated SFG and apply the same modifications to both the LBP-PUS and Shuffled LBP-PUS algorithms. For simplicity of presentation, the term “modified” is omitted in the related descriptions.

E. Complexity Reduction Using Random Update Sequences

In Algorithms 1 and 2, constructing the sorted CN set Q is required for each LBP update. However, the sorting process can be time-consuming and pose challenges for hardware implementation. Therefore, in this subsection, we investigate the impact of replacing the sorted update sequence with a random update sequence, where the update order of the CNs in Q is determined afresh at each construction, without following any fixed pseudo-random sequence. In the following, the label R is used to denote the schemes that consider random update sequences.

As demonstrated in Fig. 17, the sorting process plays an important role in improving the BER performance when considering LBP-PUS M0, which denotes the LBP-PUS algorithm that does not utilize any threshold value. The same phenomenon can be observed when considering LBP-PUS M1.

When analyzing LBP-PUS M2, recall that μ_1 and σ_r respectively denote the mean and standard deviation of the non-zero $r_{c_i, \max}$ values. We further select two μ values from

a neighborhood of μ_1 according to Fig. 9. It is noted that while we adopt the choice $\mu_2 \equiv \mu_1 + 0.1\sigma_r$ and $\mu_3 \equiv \mu_1 - 0.1\sigma_r$, other proximate values are equally admissible. In Fig. 17, it is noteworthy that the sorting process provides only limited assistance when the μ value is non-zero. Moreover, this phenomenon appears to be relatively insensitive to the μ values close to μ_1 . Given that the performance loss incurred by using a random update is minor for μ values of interest, we have chosen to adopt algorithms incorporating random updates for the remainder of this paper, as they offer the advantage of reduced computational complexity.

IV. PROPOSED BELIEF PROPAGATION-BASED POLAR DECODER WITH SCHEDULE DIVERSITY AND BIT-FLIPPING

A. Implementation of Schedule Diversity and Bit-Flipping

As defined in [20], schedule diversity is achieved by employing different message update orders in each decoding attempt, where each attempt consists of at most $I_{\max}$ iterations. For a fixed schedule such as flooding, only a single update order exists, namely, all messages are updated simultaneously. In contrast, IDS-based strategies potentially enable various update orders to be generated for the same received signal vector under a fixed code graph. Motivated by this concept, our work first identifies IDS strategies that are well suited for LDPC-like SFGs and capable of delivering fast convergence. Based on this foundation, if the first decoding attempt fails, subsequent attempts can adopt distinct message update orders. For example, instead of always selecting the CN with the largest residual value as the next node to update, one may randomly choose among the top three CNs. Owing to the fast convergence property, the current attempt can be terminated early once the stopping criterion is satisfied, and a new attempt can then be initiated. Consequently, each decoding attempt naturally produces a different update sequence, thereby enabling schedule diversity.

Algorithm 3 LBP-PUS and Shuffled LBP-PUS Decoding Algorithms With Schedule Diversity and Bit-Flipping

- 1: Store the L_j values for all j .
- 2: Set the flag PC = FALSE.
- 3: Set decoding attempt index $d = 0$ and the maximum decoding attempt number to $D_{\max}$.
- 4: Perform Algorithm 1 or 2 with random update sequences based on the stored L_j values. $d = d + 1$.
- 5: If all of the CNs are satisfied, set PC = TRUE and goto line 10.
- 6: **while** $d < D_{\max}$ and PC = FALSE **do**
- 7: Perform the bit flipping based on the d -th bit-flipping pattern. Execute Algorithm 1 or 2 with random update sequences based on the stored L_j values. $d = d + 1$.
- 8: If all of the CNs are satisfied, set PC = TRUE and goto line 10.
- 9: **end while**
- 10: Return the decoded results.

Furthermore, the analysis in Section III-E demonstrates that for the investigated algorithms, the performance degradation

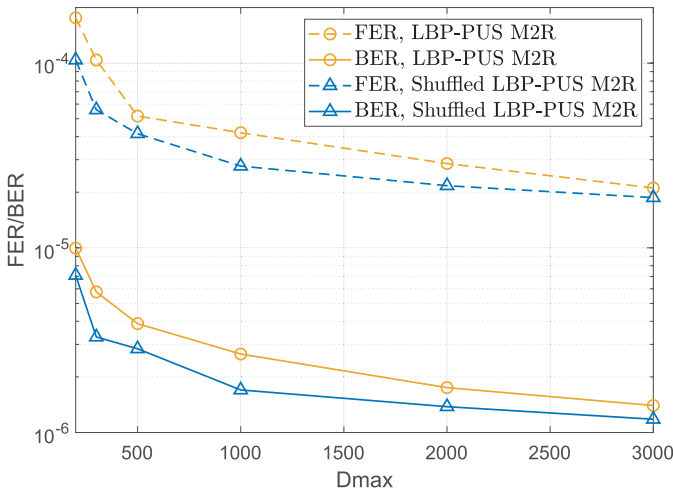


Fig. 18. Convergence performance of the proposed BP-based decoders with schedule diversity. $\omega = 1024$ and $E_b/N_0 = 2.4$ dB.

caused by random updates is negligible for the μ value of interest. This indicates that a purely random update order can be adopted, which is even better suited to a decoder designed for schedule diversity. A bit-flipping strategy is further incorporated to enhance the variation among different decoding attempts. Finally, the proposed BP-based decoding algorithm with schedule diversity is presented in Algorithm 3.

In Step 7 of Algorithm 3, the construction of bit-flipping patterns begins by selecting ω VNCHs with the smallest magnitudes of the summation of all incoming LLR values, including both channel LLR values and C2V LLR values. From these ω nodes, we iteratively modify the channel LLR values to generate flipping patterns.

Initially, we select one node at a time and replace its corresponding channel LLR value with $+\gamma$ and $-\gamma$, where γ is independent of the channel characteristics and set to a sufficiently large finite value to approximate infinity. This emulates a scenario in which the decoder has extremely high confidence in certain code-bit values. This process produces 2ω flipping patterns. Subsequently, we select two nodes at a time and modify their respective channel LLR values, resulting in an additional $4 \cdot \binom{\omega}{2}$ patterns. This pattern generation process continues for higher-order combinations as described in [25]. In total, Algorithm 3 requires up to $D_{\max} - 1$ flipping patterns (decoding attempts) to achieve the desired outcome. However, based on our experimental evidence, flipping exactly one bit per attempt yields superior error-rate performance. Accordingly, we set ω to be as large as is practically feasible, with its value determined by the code length, 1024, in the present setting. Fig. 18 presents FER and BER as functions of the maximum number of decoding attempts. The Shuffled LBP-PUS M2R scheme exhibits superior convergence in both metrics. In particular, the curves are observed to essentially converge once $D_{\max} \geq 3000$.

Channel LLR values are initially stored in memory. During the first decoding attempt, the process terminates if the stopping criterion specified in Algorithms 1 and 2 is met. To prioritize fast convergence, early termination may occur during the decoding process. If early termination does not

take place, the channel LLR values are retrieved from memory after $I_{\max}$ iterations for the next decoding attempt. Due to the adoption of random updates, the decoding schedule varies between attempts, introducing schedule diversity. Additionally, bit-flipping strategy further enhances the diversity of the decoding outcomes. This process continues until the stopping criterion is satisfied or the maximum number of attempts is reached.

B. Numerical Results

In this subsection, the error performance and computational complexity of the proposed BP-based decoders with schedule diversity are analyzed and compared to those of the CA-SCL decoder [9], and two recently introduced soft-output decoders: the Soft List decoder [12] and the Fast SOL decoder [14].

In research papers on baseband signal processing algorithms for communications, the number of Floating-Point Operations (FLOPs) is commonly adopted in the literature for evaluating computational complexity [28], [29], [30]. However, using FLOPs to compare the complexity of different decoders in this work is challenging for three main reasons. First, residual values are computed using the min-sum algorithm, while the calculation of C2V messages relies on non-linear functions, such as the hyperbolic tangent, in the SPA algorithm. Consequently, it is not straightforward to assess the complexity of the proposed decoders, which considers both residual and C2V computations. Second, SC-based decoders operate on the polar encoding graph, where the low-complexity min-sum algorithm is applied to compute C2V messages. In contrast, for the proposed BP-based decoders on LDPC-like SFGs, the SPA is originally adopted. Third, in the original SFG, a regular CN has on average 3.53 incident edges, whereas after CRC integration, a CRC CN has on average 141.75 edges. Based on the use of SPA, the per-update cost for a regular CN versus a CRC CN differs substantially.

To address these challenges, for the proposed BP-based decoders, we replace the SPA algorithm with the offset min-sum algorithm [31] and properly adjust the offset values according to the SNR and CN degrees. It has been verified that the error performance curve under SPA can be closely approximated by that under the offset min-sum algorithm. Consequently, the complexity associated with residual and C2V calculations becomes comparable, and the update cost per CN is now closely aligned between SC-based and BP-based decoders. Moreover, the complexity of C2V updates becomes comparable across regular and CRC CNs in the SFG. Therefore, we ultimately adopt the number of CN updates as a fair complexity metric in this section. It is worth noting that, for the proposed decoders, the residual calculations and CRC processing are already included in the CN update count.

In the following, we adopt FER as the primary performance metric, since it is more informative than BER in scenarios where successful decoding is determined at the frame/packet level (e.g., CRC-validated decoding). Furthermore, to clearly isolate and demonstrate the respective contributions of schedule diversity and the bit-flipping strategy to the FER performance, we conduct the following two benchmark experiments:

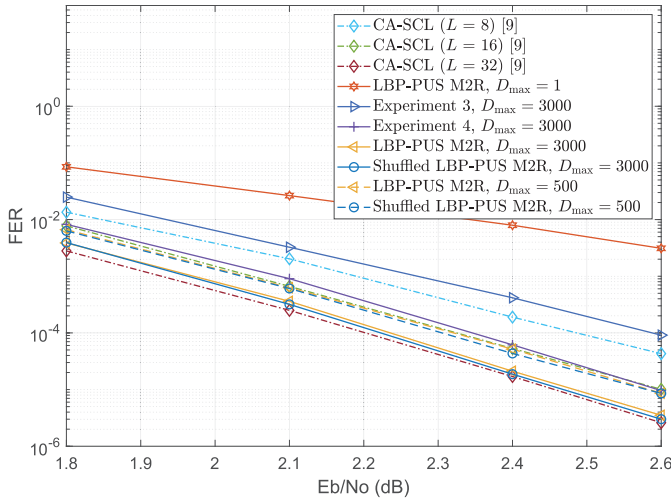


Fig. 19. Comparison of FER performance between the CA-SCL decoder and the proposed BP-based decoders with schedule diversity. $I_{\max} = 80$. $\omega = 1024$.

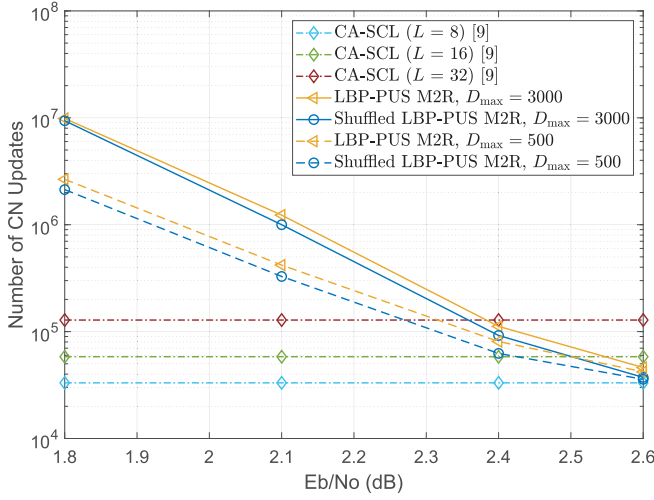


Fig. 20. Comparison of computational performance between the CA-SCL decoder and the proposed BP-based decoders with schedule diversity. $I_{\max} = 80$. $\omega = 1024$.

Experiment 3: We perform at most $D_{\max}$ decoding attempts, each using a distinct bit-flipping pattern and allowing up to $I_{\max}$ iterations. However, for every decoding attempt, the flooding schedule is employed, i.e., no schedule diversity is applied.

Experiment 4: We perform at most $D_{\max}$ decoding attempts using Algorithm 3 with LBP-PUS M2R scheduling, each consisting of up to $I_{\max}$ iterations. However, all decoding attempts start from the same set of LLR values, i.e., no bit-flipping is applied.

From Fig. 19, when Algorithm 3 is applied based on the LBP-PUS M2R algorithm with $D_{\max} = 1$, the resulting scheme is identical to the one discussed in Section III-E. Under Experiment 4, however, the corresponding FER curve approaches that of the CA-SCL decoder with $L = 16$, demonstrating that schedule diversity is successfully achieved through the investigated IDS strategies. In contrast, Fig. 19 also shows that under Experiment 3, executing multiple decoding attempts solely based on different bit-flipping patterns

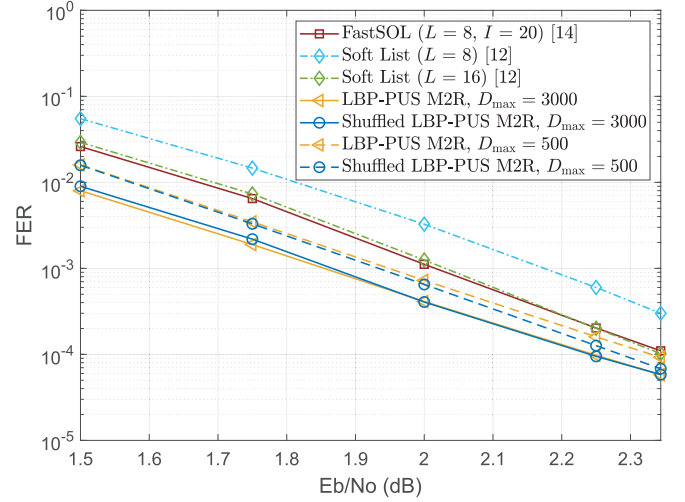


Fig. 21. Comparison of BER performance among the FastSOL decoder, the Soft List decoder, and the proposed BP-based decoders with schedule diversity. $I_{\max} = 80$. $\omega = 1024$.

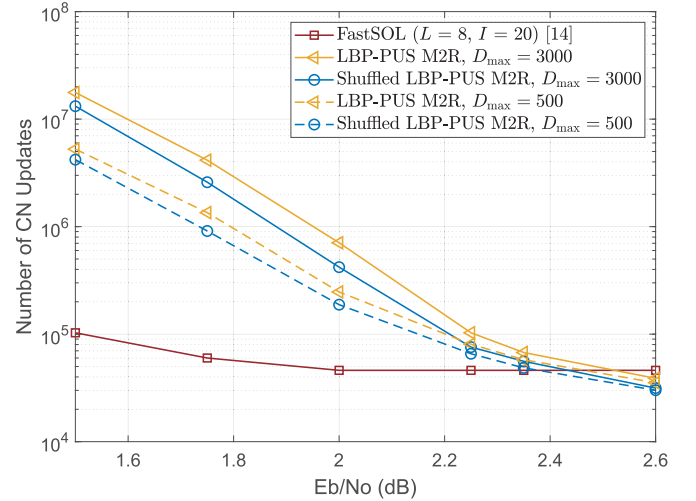


Fig. 22. Comparison of computational complexity between the FastSOL decoder and the proposed BP-based decoders with schedule diversity. $I_{\max} = 80$. $\omega = 1024$.

improves FER performance, yet the gain remains smaller than that obtained under Experiment 4. Consequently, when the combination of schedule diversity and bit-flipping strategy, i.e., Algorithm 3, is employed, the proposed decoders based on the LBP-PUS M2R and Shuffled LBP-PUS M2R algorithms with $D_{\max} = 3000$ achieve FER performance comparable to that of the CA-SCL decoder with $L = 32$. Likewise, the decoders using the LBP-PUS M2R and Shuffle LBP-PUS M2R algorithms with $D_{\max} = 500$ deliver performance comparable to that of the CA-SCL decoder with $L = 16$.

The computational complexity comparison is illustrated in Fig. 20. Considering that coded communication systems typically operate in regions where the FER is lower than 10^{-5} , a comparison between the proposed decoders with $D_{\max} = 3000$ and the CA-SCL decoder with $L = 32$ value exceeds 2.45 dB. Furthermore, Fig. 20 illustrates that the proposed decoders with $D_{\max} = 3000$ achieve lower

computational complexity compared to the CA-SCL decoder with $L = 32$ under the same condition ($E_b/N_0 > 2.45$ dB).

In Figs. 21 and 22, we compare the FastSOL decoder, the Soft List decoder, and the proposed decoders. Following the approach outlined in [14], we further consider the polar code specified in the 5G standard [32], where the CRC length is 11 and the resultant rate is $R = 501/1024$. The proposed decoders, based on the LBP-PUS M2R and Shuffle LBP-PUS M2R algorithms with $D_{\max} = 3000$, achieve a performance gain of approximately 0.1 dB over the FastSOL ($L = 8, I = 20$) and Soft List ($L = 16$) decoders at a FER of 10^{-4} . Similarly, as shown in Fig. 22, the proposed decoders exhibit a decreasing complexity requirement, and are comparable to the FastSOL decoder in the high-SNR region.

V. CONCLUSION

In this paper, novel BP-based polar decoders are proposed, leveraging the concept of schedule diversity. Observing the less favorable convergence performance and suboptimal BER results of conventional IDS strategies when applied to the LDPC-like SFG, the impact of VNHs is analyzed, and appropriate scheduling algorithms are developed by integrating both SSS and IDS strategies. Systematic polar codes are employed to enable the incorporation of CRC bits into the SFG, thereby enhancing error performance. Additionally, random update sequences are introduced to mitigate the hardware burden and facilitate the implementation of schedule diversity. Numerical results demonstrate that the proposed BP-based polar decoders with schedule diversity achieve comparable or improved error performance while reducing computational complexity in the high SNR region compared to benchmark decoders. These results serve as preliminary findings on the BP algorithm over LDPC-like SFGs and provide useful insights for the design and practical improvement of polar decoders.

REFERENCES

- [1] C. Berrou, A. Glavieux, and P. Thitimajshima, "Near Shannon limit error-correcting coding and decoding: Turbo-codes. 1," in *Proc. IEEE Int. Conf. Commun.*, vol. 2, May 1993, pp. 1064–1070.
- [2] Y. Hawwar et al., "3G UMTS wireless system physical layer: Baseband processing hardware implementation perspective," *IEEE Commun. Mag.*, vol. 44, no. 9, pp. 52–58, Sep. 2006.
- [3] Z. Shen, A. Papasakellariou, J. Montojo, D. Gerstenberger, and F. Xu, "Overview of 3GPP LTE-advanced carrier aggregation for 4G wireless communications," *IEEE Commun. Mag.*, vol. 50, no. 2, pp. 122–130, Feb. 2012.
- [4] D. J. MacKay and R. M. Neal, "Near Shannon limit performance of low density parity check codes," *Electron. Lett.*, vol. 33, no. 6, pp. 457–458, Mar. 1997.
- [5] K. Teo, Z. Tao, and J. Zhang, "The mobile broadband WiMAX standard [standards in a nutshell]," *IEEE Signal Process. Mag.*, vol. 24, no. 5, pp. 144–148, Sep. 2007.
- [6] ETSI, *Digital Video Broadcasting (DVB); Second Generation Framing Structure, Channel Coding and Modulation Systems for Broadcasting, Interactive Services, News Gathering and Other Broadband Satellite Applications*, Part II: S2-Extensions (DVB-S2X), ETSI, Sophia Antipolis, France, 2005, pp. 22–27.
- [7] E. Arıkan, "Channel polarization: A method for constructing capacity-achieving codes for symmetric binary-input memoryless channels," *IEEE Trans. Inf. Theory*, vol. 55, no. 7, pp. 3051–3073, Jul. 2009.
- [8] *Chairmans Notes of Agenda Item 7.1.5 Channel Coding and Modulation*, document 3GPP TSG RAN WG1 Meeting #87, R 1-1613710, Lisbon, Portugal, Oct. 2016.
- [9] I. Tal and A. Vardy, "List decoding of polar codes," *IEEE Trans. Inf. Theory*, vol. 61, no. 5, pp. 2213–2226, May 2015.
- [10] K. Niu and K. Chen, "CRC-aided decoding of polar codes," *IEEE Commun. Lett.*, vol. 16, no. 10, pp. 1668–1671, Oct. 2012.
- [11] A. Balatsoukas-Stimming, P. Giard, and A. Burg, "Comparison of polar decoders with existing low-density parity-check and turbo decoders," in *Proc. IEEE Wireless Commun. Netw. Conf. Workshops (WCNCW)*, Mar. 2017, pp. 1–6.
- [12] L. Xiang, Y. Liu, Z. B. K. Egilmez, R. G. Maunder, L.-L. Yang, and L. Hanzo, "Soft list decoding of polar codes," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13921–13926, Nov. 2020.
- [13] U. U. Fayyaz and J. R. Barry, "Low-complexity soft-output decoding of polar codes," *IEEE J. Sel. Areas Commun.*, vol. 32, no. 5, pp. 958–966, May 2014.
- [14] Y. Shen, W. Zhou, Y. Huang, Z. Zhang, X. You, and C. Zhang, "Fast iterative soft-output list decoding of polar codes," *IEEE Trans. Signal Process.*, vol. 70, pp. 1361–1376, 2022.
- [15] E. Arıkan, "Polar codes: A pipelined implementation," in *Proc. 4th Int. Symp. Broad. Commun. (ISBC)*, 2010, pp. 11–14.
- [16] N. Goela, S. B. Korada, and M. Gastpar, "On LP decoding of polar codes," in *Proc. IEEE Inf. Theory Workshop*, Aug. 2010, pp. 1–5.
- [17] S. Cammerer, M. Ebada, A. Elkelesh, and S. ten Brink, "Sparse graphs for belief propagation decoding of polar codes," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Vail, CO, USA, Jun. 2018, pp. 1465–1469.
- [18] A. Elkelesh, M. Ebada, S. Cammerer, and S. ten Brink, "Belief propagation list decoding of polar codes," *IEEE Commun. Lett.*, vol. 22, no. 8, pp. 1536–1539, Aug. 2018.
- [19] D. E. Hocevar, "A reduced complexity decoder architecture via layered decoding of LDPC codes," in *Proc. IEEE Workshop Signal Process. Syst. (SIPS)*, Oct. 2004, pp. 107–112.
- [20] H.-C. Lee and Y.-L. Ueng, "LDPC decoding scheduling for faster convergence and lower error floor," *IEEE Trans. Commun.*, vol. 62, no. 9, pp. 3104–3113, Sep. 2014.
- [21] G. Elidan, I. McGraw, and D. Koller, "Residual belief propagation: Informed scheduling for asynchronous message passing," in *Proc. 22nd Conf. Uncertainty Artif. Intell. (UAI)*, Cambridge, MA, USA, Jul. 2006, pp. 165–173.
- [22] A. I. V. Casado, M. Griot, and R. D. Wesel, "Informed dynamic scheduling for belief-propagation decoding of LDPC codes," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2007, pp. 932–937.
- [23] J.-H. Kim, M.-Y. Nam, H.-Y. Song, and K.-M. Lee, "Variable-to-check residual belief propagation for informed dynamic scheduling of LDPC codes," in *Proc. Int. Symp. Inf. Theory Appl.*, Auckland, New Zealand, Dec. 2008, pp. 1–4.
- [24] H.-C. Lee, Y.-L. Ueng, S.-M. Yeh, and W.-Y. Weng, "Two informed dynamic scheduling strategies for iterative LDPC decoders," *IEEE Trans. Commun.*, vol. 61, no. 3, pp. 886–896, Mar. 2013.
- [25] Y. Yu, Z. Pan, N. Liu, and X. You, "Belief propagation bit-flip decoder for polar codes," *IEEE Access*, vol. 7, pp. 10937–10946, 2019.
- [26] E. Arıkan, "Systematic polar coding," *IEEE Commun. Lett.*, vol. 15, no. 8, pp. 860–862, Aug. 2011.
- [27] H.-F. Yu, H.-C. Lee, and S.-K. Lee, "Early termination belief propagation decoder with parity check matrix for polar codes," in *Proc. 27th Wireless Opt. Commun. Conf. (WOCC)*, Taiwan, Apr. 2018, pp. 1–4.
- [28] A. Mohammad, C. Masouros, and Y. Andreopoulos, "Complexity-scalable neural-network-based MIMO detection with learnable weight scaling," *IEEE Trans. Commun.*, vol. 68, no. 10, pp. 6101–6113, Oct. 2020.
- [29] M. Arvinte and J. I. Tamir, "MIMO channel estimation using score-based generative models," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3698–3713, Jun. 2023.
- [30] S. Bhattacharya, K. V. S. Hari, and Y. C. Eldar, "Unfolding for symbol detection in task-based quantized MIMO receivers," *IEEE Wireless Commun. Lett.*, vol. 14, no. 3, pp. 926–930, Mar. 2025.
- [31] X. Wu, Y. Song, M. Jiang, and C. Zhao, "Adaptive-normalized/offset min-sum algorithm," *IEEE Commun. Lett.*, vol. 14, no. 7, pp. 667–669, Jul. 2010.
- [32] 5G; NR; *Multiplexing and Channel Coding*, Release 16, V16.2.0, document 3GPP TS 38.212, Jul. 2020. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/138200_138299/138212/16.02.00_60/ts_138212v160200p.pdf



Cheng-Jun You received the B.S. degree in electronic engineering from the National Changhua University of Education, Changhua, Taiwan, in 2022, and the M.S. degree in communications engineering from National Sun Yat-sen University, Kaohsiung, Taiwan, in 2024. His research interests include wireless communications, deep learning, and error-correcting codes.



Huang-Chang Lee (Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from National Chi Nan University, Taiwan, in 2002 and 2004, respectively, and the Ph.D. degree in electrical engineering from National Tsing Hua University in 2014. He is currently an Associate Professor with the Department of Electrical Engineering, National Taiwan Ocean University, Keelung, Taiwan. His research interests include channel coding, digital signal processing, and their applications to communications.



Yi-Hsin Chen received the B.S. degree in electronic engineering from the National Changhua University of Education, Changhua, Taiwan, in 2021, and the M.S. degree in communications engineering from National Sun Yat-sen University, Kaohsiung, Taiwan, in 2023. She is currently a Baseband Software Engineer with the Industrial Technology Research Institute, Taiwan. Her research interests include wireless communications, high-PHY signal processing, and error-correcting codes.



Chen-Wei Hu received the B.S. degree in electronics engineering from National Chi Nan University, Nantou, Taiwan, in 2019, and the M.S. degree in communications engineering from National Sun Yat-sen University, Kaohsiung, Taiwan, in 2021. He is currently a Firmware Engineer with Electronics Industry. His research interests include wireless communications, MIMO-OFDM signal processing, and error-correcting codes.



Yen-Ming Chen (Member, IEEE) received the Ph.D. degree in electrical engineering from National Tsing Hua University (NTHU), Hsinchu, Taiwan, in 2011. From 2011 to 2014, he was a Post-Doctoral Research Fellow with the Department of Electrical Engineering, NTHU. He was an Assistant Professor with the Department of Electronic Engineering, Chung Yuan Christian University, Taoyuan, Taiwan, from 2014 to 2016. From 2017 to 2025, he was with National Sun Yat-sen University, Kaohsiung, Taiwan, where he held a faculty position with the Institute of Communications Engineering and a Full Professor. Since 2025, he has been with the Institute of Communications Engineering, National Yang Ming Chiao Tung University, Hsinchu, where he is currently an Associate Professor. His research interests include error-correcting codes, coded modulation, non-orthogonal multiple access, and biometrics. In 2020, he was a recipient of the Best Paper Award for Young Scholars from the IEEE IT/COM Society Taipei/Tainan Chapters, and also the Best Young Professional Member Award from the IEEE Tainan Section.